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Studierichting: Informatica
Oefeningenreeks 2E nr. 26.34

$(\log_3 x)^2 - 3^{\log_3 10} = \log_3 x^6 - 9^{\log_3 \sqrt{3}}$

$\Leftrightarrow (\log_3 x)^2 - 10 = 6 \log_3 x - 3^{2 \log_3 \sqrt{3}}$

$\Leftrightarrow (\log_3 x)^2 - 6 \log_3 x - 10 = -3$

$\Leftrightarrow (\log_3 x)^2 - 6 \log_3 x - 7 = 0$ stel $t = \log_3 x$

$\Leftrightarrow t^2 - 6t - 7 = 0$

$\Delta = 64$

$t_{1,2} = \dfrac{6 \pm 8}{2}$

$t_1 = 7$

$t_2 = -1$

dus $x = -\dfrac{1}{3} \vee x = 2187$

References